

Year 10
Pathway 1

Mini Log Test

Name: Solutions

Class: _____

Time: 20 minutes

Total Mark: ____/34

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning will be allocated zero marks.

Resources: $\frac{1}{2}$ A4 page of notes (1 side), Calculator allowed.

Question 1

3 marks

Write the following in log form

a. $5^x = 125$

$\log_5 125 = x$

b. $4^2 = 16$

$\log_4 16 = 2$

c. $3^4 = 81$

$\log_3 81 = 4$

Question 2

3 marks

Write the following in index form

a. $\log_4 64 = x$

$4^x = 64$

b. $\log_8 8 = 1$

$8^1 = 8$

c. $\frac{1}{2} = \log_{49} 7$

$49^{\frac{1}{2}} = 7$

Question 3

3 marks

Evaluate

a. $\log_2 16 = 4$

b. $\log_{11} 121 = 2$

c. $\log_2 128 = 7$

Question 4

(2,2,2,2,3) 11 marks

I. Evaluate the following

a. $\log_6 3 + \log_6 2$

$$\begin{aligned} & \log_6 3 \times 2 \\ &= \log_6 6 \\ &= 1 \end{aligned}$$

c. $\log_{10} 25 + \log_{10} 4$

$$\begin{aligned} & \log_{10} 25 \times 4 \\ &= \log_{10} 100 = 2 \end{aligned}$$

e. $\log_3 27 + \log_3 2 - \log_3 6$

$$\begin{aligned} &= \log_3 \frac{27 \times 2}{6} \\ &= \log_3 9 = 2 \end{aligned}$$

b. $\log_2 20 - \log_2 5$

$$\begin{aligned} &= \log_2 20 \div 5 \\ &= \log_2 4 \\ &= 2 \end{aligned}$$

d. $\log_{10} 30000 - \log_{10} 3$

$$\begin{aligned} & \log_{10} 30000 \div 3 \\ & \log_{10} 10000 = 4 \end{aligned}$$

Question 5

(2,3,3) 8 marks

Solve the following.

a. $\log_2 x = 3$

$$\begin{aligned} & 2^3 = x \\ & \therefore x = 8 \end{aligned}$$

c. $\log_5 (x - 1) = 2$

$$\begin{aligned} & 5^2 = x - 1 \\ & 25 = x - 1 \\ & \begin{array}{r} + 1 \quad + 1 \\ \hline 26 = x \end{array} \\ & \therefore x = 26 \end{aligned}$$

b. $\log_3 x^4 = -16$

$$\begin{aligned} & 4 \log_3 x = -16 \\ & \log_3 x = -16 \div 4 \\ & \log_3 x = -4 \\ & 3^{-4} = x \\ & \therefore x = \frac{1}{81} \end{aligned}$$

If $\log 3.75 = 0.574031$, solve

a. $\log 3750$

$$\begin{aligned}\log 3.75 \times 1000^{\frac{1}{10}} \\ \log 3.75 + \log 1000 \\ 0.574031 + 3 \\ = 3.574031\end{aligned}$$

b. $\log 0.000375$

$$\begin{aligned}\log 3.75 \times 10^{-4} \\ 0.574031 - 4 \\ = -3.425969\end{aligned}$$